

BHARAT MISHRA

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EDUCATION

- **Bachelor of Technology, VIT Bhopal University** | Computer Science Engineering | **CGPA: 9.02** [2023 - 2027]
- **Grade XII, Vidyatree Modern World School** | Percentage: 95.50 [2021 - 2023]

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Java, SQL
- **ML / AI:** TensorFlow, Scikit-learn, FastAPI, Pandas, NumPy, OpenAI Gym
- **Developer Tools:** GitHub, Git, MySQL, Docker, PostgreSQL
- **Core Concepts:** Neural Networks, Reinforcement Learning, Embeddings, Recommendation Systems, Cosine Similarity Search, NLP & LLM Pipelines

AI & MACHINE LEARNING PROJECTS

- **DepoIndex: AI-Powered Legal Document Indexing**
 - Engineered an AI-powered tool to automate the generation of a table of contents from legal deposition PDFs, significantly reducing the manual effort required for **document analysis**.
 - Designed an efficient **preprocessing pipeline** that annotates text with precise page and line numbers, minimizing prompt token count while enabling the LLM to extract topics with accurate source references.
 - Achieved **95% page-level and 85% line-level accuracy** in topic referencing through manual validation, delivering a highly reliable tool for automated document indexing.
- **MovieMix: Latent Semantic Movie Search Engine (MovieLens 25M)**
 - Built a **semantic** search system that converts raw user input tags into 100-D **embeddings**, and retrieves movies via **vectorized cosine similarity against MF-derived movie embeddings**
 - Engineered a **500-token tag vocabulary** and a hypersphere-normalized text-to-vector neural network for solving the **hubness problem** and also fully solving the **cold-start problem** by generating embeddings from tags alone, that achieved **0.52 cosine similarity**.
 - Deployed a fast <50ms **inference pipeline** using **FastAPI + Docker**, enabling production-grade semantic movie retrieval.
- **Search and Rescue Reinforcement Learning System**
 - Developed autonomous agents using Q-learning and PPO to optimize pathfinding in a dynamic, multi-agent search and rescue simulation.
 - **Engineered the simulation environment and its core physics from scratch**, defining the complete state space, action space, and reward functions for the RL agents.
 - Achieved an approximate **90% reduction in target search time** for the trained agent compared to a random-walk baseline.
- **Assignment Management System (Python, PostgreSQL) (In Progress)**
 - Building a backend system to manage users, assignments, and submissions using Python with PostgreSQL.
 - Designed a normalized relational database schema with primary and foreign keys to ensure data consistency.
 - Using **Pydantic** models for input validation and structured data handling, and implementing core database operations with clean separation between application logic and persistence.

ACHIEVEMENTS & CERTIFICATIONS

- **Global Rank 8261, TCS CodeVita (2024):** Advanced to the penultimate round by solving complex algorithmic problems.
- **Generative AI Training (Kaggle/Google) (2024):** Transformers, LLMs, and prompt engineering.
- **Google Certified: Computer Networking:** The Bits and Bytes of Computer Networking.

LEADERSHIP AND VOLUNTEERING

- **PR and Outreach Lead, Startup Club:** Secured industry partnerships for guest lectures and sponsorships by coordinating directly with C-suite executives and other industry professionals.
- **Volunteer, Pahal Foundation:** Taught basic mathematics and fundamental elementary education to underprivileged children.